Ethics & AI: AI alignment

Jean Mark Gawron

San Diego State

May 8, 2026

Table of contents

Prologue: From The Atlantic (April 2026) *Insatiable* by Matteo Wong

Colossus

...Finally, I saw it — a white-walled hangar bigger than a dozen football fields, where Elon Musk intends to build a god.
This is **Colossus**: a data-center Musk's artificial intelligence company, xAI, is using as a training ground for **Grok**, one of the world's most advanced generative AI models. Training these models takes a staggering amount of energy; if run at full strength for a year, Colussus would use as much energy as 200,000 American homes. When fully operational [Musk said] this facility and two other xAI data centers nearby will require **nearly two gigaWatts of power. Annually, those facilities could consume roughly twice as much electricity as the city of Seattle.**

You can't make this stuff up

Well, no, actually it already has been made up . . .

Colossus
: Colossus is a 1966 science fiction novel by British author D. F. Jones which explores the dangers of artificial intelligence. Shortly after activation, Colossus exceeds expectations and becomes self-aware. It demands to be connected to “Guardian," its Soviet counterpart, leading the two computers to take control of human affairs to prevent war by removing human agency.

Grok
: a neologism coined by American author Robert A. Heinlein in his 1961 sci-fi novel *Stranger in a Strange Land*, meaning to understand something so thoroughly, intuitively, or empathically that the observer becomes part of the observed. It implies deep, profound comprehension — knowing something completely, rather than just intellectually.

Or the people "behind" it all

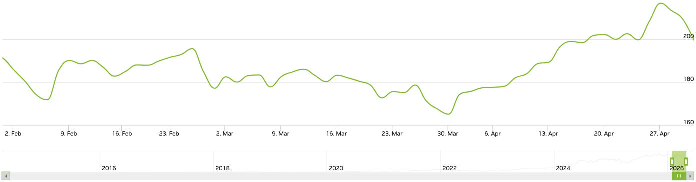
Delos D. Harriman	A ruthless and brilliant tycoon in Robert Heinlein's story "The Man who sold the Moon". The story follows Harriman as he uses political maneuvering, financial schemes, and sheer will to overcome obstacles, ultimately making space travel a reality, a narrative that reflects modern private space ventures.
Jubal Harshaw	Rich benefactor & patron saint of the religion founded in *Stranger in Stranger Land*: "a devout and fierce individualist in a world filled with cults and bureaucracies" (Dan Schneider).

The true driving force

- Because figures like Elon Musk and OpenAI founder Sam Altman and philosophies like Effective Altruism play such a prominent role in the social/financial/political history of AI, it's tempting to think that the driving force behind Generative AI is a cabal of oddball billionaires with the collective maturity of a 13-year old science fiction fan.
- But the truth of the matter is that if it wasn't Musk or Altman, it would be someone else, because the real driving force of this revolution is private profit and government control.
- The discussion to have now is not how to dismiss it, or even how to stop it; it's how not to do something stupid ("Connect Colossus up to Guardian? Sure! What could go wrong?")

Money

NVidia



xAI

xAI is a private company and does not have a publicly traded stock price;

OpenAI

OpenAI seems to be planning an IPO, but so far is selling stock to accredited investors through secondary markets.

AI Alignment

Research field

AI alignment is the research field dedicated to ensuring artificial intelligence (AI) systems' goals, behaviors, and actions match human intentions, values, and ethical principles

EU AI Act I

IBM's EU AI Act page

- **Risk Classification**: AI systems are divided into four categories: Unacceptable (banned), High-Risk (regulated), Limited Risk (transparency obligations), and Minimal Risk.
- **High-Risk Systems**: These require strict compliance, including risk management, data governance, technical documentation, and human oversight. Examples include AI in education, employment, and critical infrastructure.
- **General-Purpose AI (GPAI)**: The act sets rules for GPAI models, such as Meta's Llama 3, requiring technical documentation and compliance with copyright laws.
- **Penalties**: Fines for non-compliance go up to €35 million or 7% of total global turnover.

High Risk (EU AI Act) I

These systems are aallowed but have very specific restructions imposed on them:

- in employment contexts, such as those used to recruit candidates, evaluate applicants and make promotion decisions
- in certain medical devices
- in certain education and vocational training contexts
- in the judicial and democratic process such as systems that are intended to influence the outcome of elections
- to determine access to essential private or public services, including systems that assess eligibility for public benefits and evaluate credit scores.
- in critical infrastructure management (for example, water, gas and electricity supplies and so on)

High Risk (EU AI Act) II

- in any biometric identification system which are not prohibited, except for systems that whose sole purpose is to verify a person's identity (for example, using a fingerprint scanner to grant someone access to a banking app).

Forbidden (EU AI Act) I

From the IBM EU AI site

- **Social scoring systems**: systems that evaluate or classify individuals based on their social behavior — leading to detrimental or unfavorable treatment in social contexts unrelated to the original data collection and unjustified or disproportionate to the gravity of the behavior;
- **Emotion recognition systems**: at work and in educational institutions, except where these tools are used for medical or safety purposes;
- **Exploitation systems**: AI systems used to exploit people's vulnerabilities (for example vulnerabilities due to age or disability).
- **Data collection of facial images**: Untargeted scraping of facial images from the internet or closed-circuit television (CCTV) for facial recognition databases;

Forbidden (EU AI Act) II

- **Biometric identification systems**: systems that identify individuals based on sensitive characteristics;
- **Specific predictive policing applications**
- **Biometric identification systems**: Law enforcement use of real-time remote biometric identification systems in public (unless an exception applies, and pre-authorization by a judicial or independent administrative authority is generally required).

Transparency (EU AI Act)

From the IBM EU AI Act Page

- AI systems intended to directly interact with individuals should be designed to inform users that they are interacting with an AI system, unless this is obvious to the individual from the context. A chatbot, for example, should be designed to notify users that it is a chatbot.
- AI systems that generate text, images or other certain other content must use machine-readable formats to mark outputs as AI generated or manipulated. This includes, for example, AI that generates deepfakesimages or video altered to show someone doing or saying something they didnt do or say.

US AI Regulation I

- No single comprehensive AI Act comparable to the EU AI Act (It is the policy of the United States to sustain and enhance the United States global AI dominance through a minimally burdensome national policy framework for AI.)
- CREATE AI Act(2025): Reintroduced to establish the National Artificial Intelligence Research Resource (NAIRR) to provide researchers and students with access to data, computing power, and tools.
- Draft Legislation in the 119th Congress
  1. HEALTH AI Act
  2. the AWARE Act (chatbot safety)
  3. PILLAR Act (cyber security)
- California: Transparency in Frontier Artificial Intelligence Act and AB 2013 regarding dataset disclosure

US AI Regulation II

1. Under consideration: require insurance companies to report the use of AI when denying healthcare claims
2. Require the AI performance-testing before implementing on certain applications

Bibliography

A final note on the environmental impact

1. According to the Atlantic article cited above, the consensus in the data-center building community is that sustainable power sources like wind, solar, water, aren't up to the challenge of supplying the power needs for modern data centers.
2. So data centers like Colussus are building their own power plants.
3. Power sources like natural gas, coal, and nuclear are back in center frame as the most relaiable and efficient ways to supply power for these power gluttons.
4. Hybrid systems combining different sources are more expensive and complex to build and they don't seem to be on the table as of now.